Assignment 1: Basics

Empirical Machine Learning

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Assignment 1: Basics — empirical machine learning

Kesavan Ramakrishnan

This submission studies the course d8 language model through controlled hyperparameter, scaling, stochasticity, and diagnostic experiments. Unless stated otherwise, the standard run uses 8 layers of width 512, 614.4M fresh training tokens, AdamW, learning rate 0.003, 1% warmup, linear decay, batch size 64, weight decay 0.1, and no dropout.

Reports

- Problem 1: Hyperparameter sweeps
- Problem 2: Reliability of scaling laws
- Problem 3: Measuring variation
- Problem 4: Amplification of randomness
- Problem 5: Loss-curve augury
- Problem 6: Activations and gradient norms
- Completed-run index

Main conclusions

Learning rate and batch size have the largest direct effects on d8 performance, while warmup and weight decay matter mainly through interactions. Model-size scaling is highly predictable when the data budget is sufficient, but repeated or insufficient data breaks the trend. Ordinary seed-to-seed validation variation is a few thousandths of a nat, and tiny deterministic perturbations can produce differences of that same order. Loss-curve shape separates schedule and learning-rate problems from batch noise, while internal RMS statistics show residual growth with depth, large mid-training activations, and small late gradients.

Problem 1: Hyperparameter sweeps

All 55 runs use the d8 model and 600,000 sequences (614.4M tokens), so model size and token count are fixed. Unless named otherwise, the recipe is AdamW with learning rate 0.003, 1% warmup, linear decay, batch size 64, and weight decay 0.1.

(a) One variable at a time

Each sweep spans roughly one order of magnitude around the standard value. The entries below are final validation losses.

Variable	Settings	Final validation loss
Learning rate	0.000333, 0.001, 0.003, 0.009, 0.027	3.1265, 2.9777, **2.9259**, 2.9493, 3.0172
Warmup fraction	0.001, 0.003, 0.01, 0.03, 0.10	2.9945, 2.9402, 2.9259, 2.9245, **2.9202**

Variable	Settings	Final validation loss
Batch size	16, 32, 64, 128, 256	2.9272, **2.9185**, 2.9259, 2.9497, 3.0136
Weight decay	0.01, 0.03, 0.10, 0.30, 1.0	2.9466, 2.9420, 2.9259, **2.9200**, 2.9624

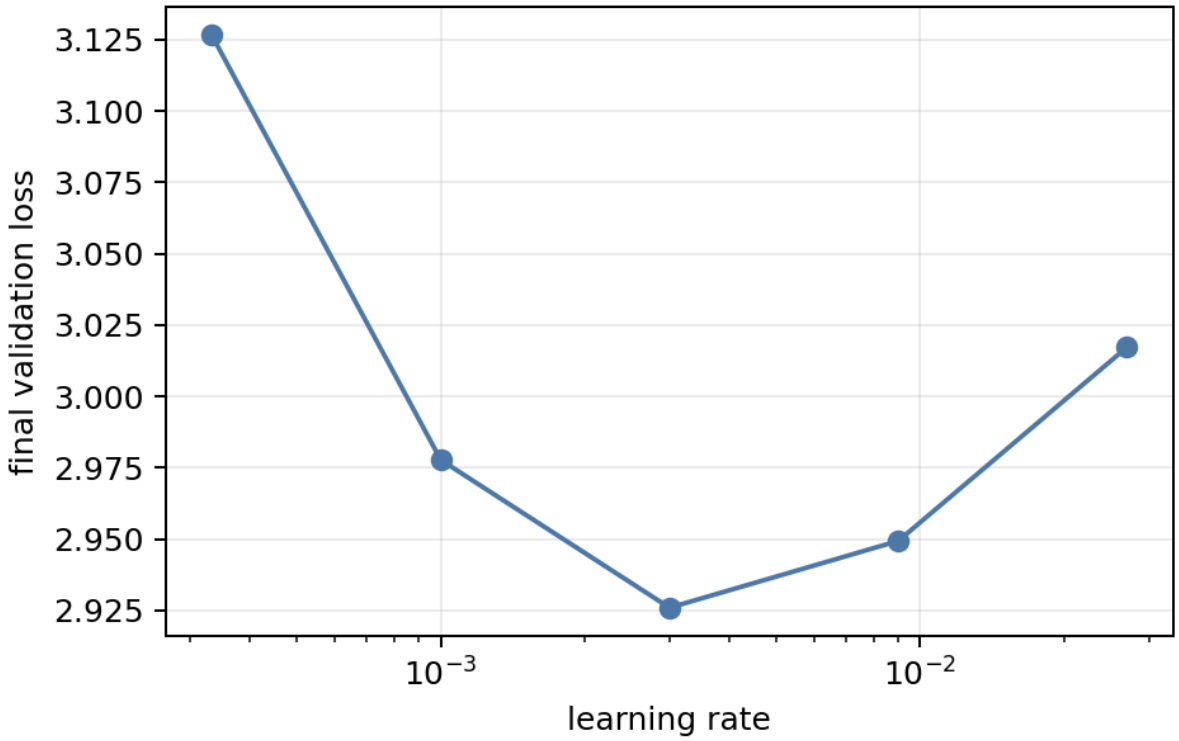


Figure 1: Learning-rate sweep

Learning rate has a clear optimum near 0.003. Warmup and weight decay are comparatively flat near their optima: 3–10% warmup and weight decay 0.1–0.3 all work well. Batch size 32 is best in this sweep; performance degrades sharply at 128 and 256. The important effect sizes are about 0.20 loss across the learning-rate sweep and 0.10 across the batch-size sweep, compared with only 0.04 across the useful warmup and weight-decay ranges.

(b) Interactions

The pairwise experiments vary learning rate with batch size, warmup, or weight decay while holding the remaining settings fixed.

Learning rate 0.003 remains best at every fully compared batch size. At batch 32, losses for learning rates 0.001, 0.003, 0.009, and 0.027 are 2.9429, 2.9185, 2.9514, and 3.0276. At batch 256 they are 3.1051, 3.0136, 3.1046, and 3.1350. Larger batches therefore worsen the attainable loss without moving the optimum away from 0.003.

Warmup must be tuned with learning rate. Increasing warmup from 0.01 to 0.10 improves learning rate 0.009 from 2.9493 to 2.9135, making it better than the standard run. The same change improves learning rate 0.027 from 3.0172 to 2.9588, although it remains worse than the optimum.

Weight decay also interacts strongly with learning rate. At weight decay 1.0, learning rate 0.001 gives 2.9326, while 0.003, 0.009, and 0.027 give 2.9624, 3.1599, and 3.3343. Strong decay therefore shifts the preferred learning rate downward. The best observed pair is learning rate 0.003 with weight decay 0.3, at 2.9200.

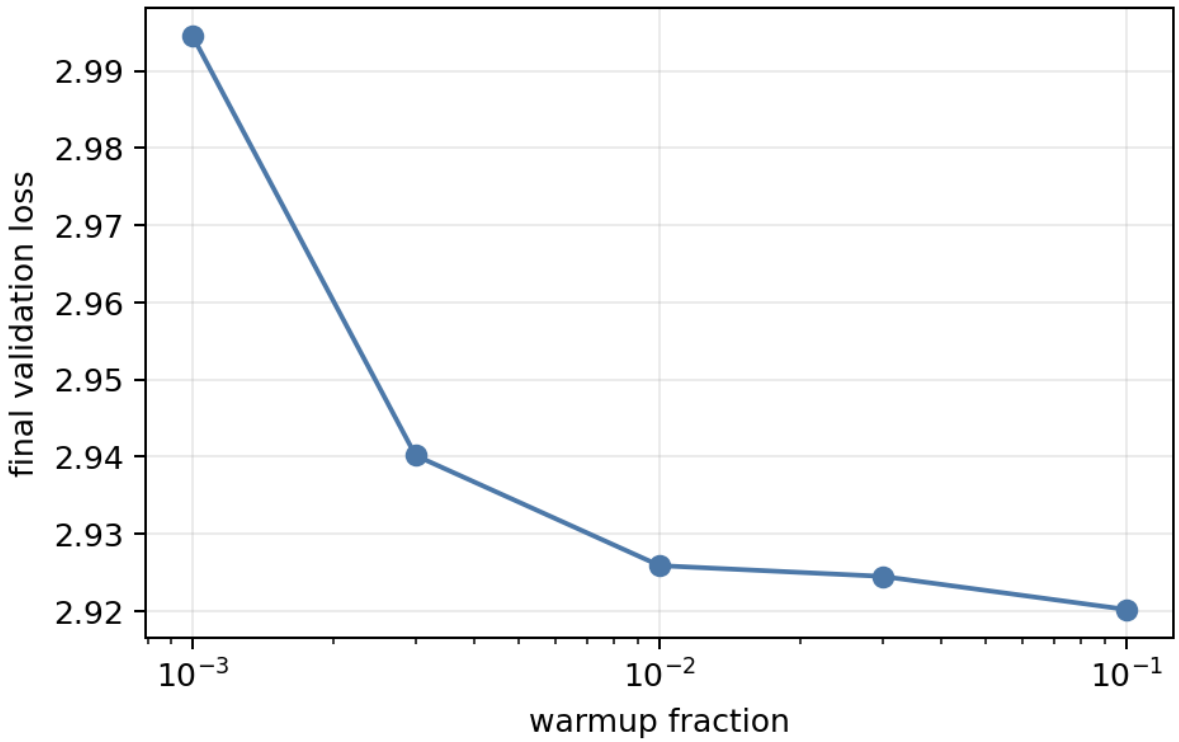


Figure 2: Warmup sweep

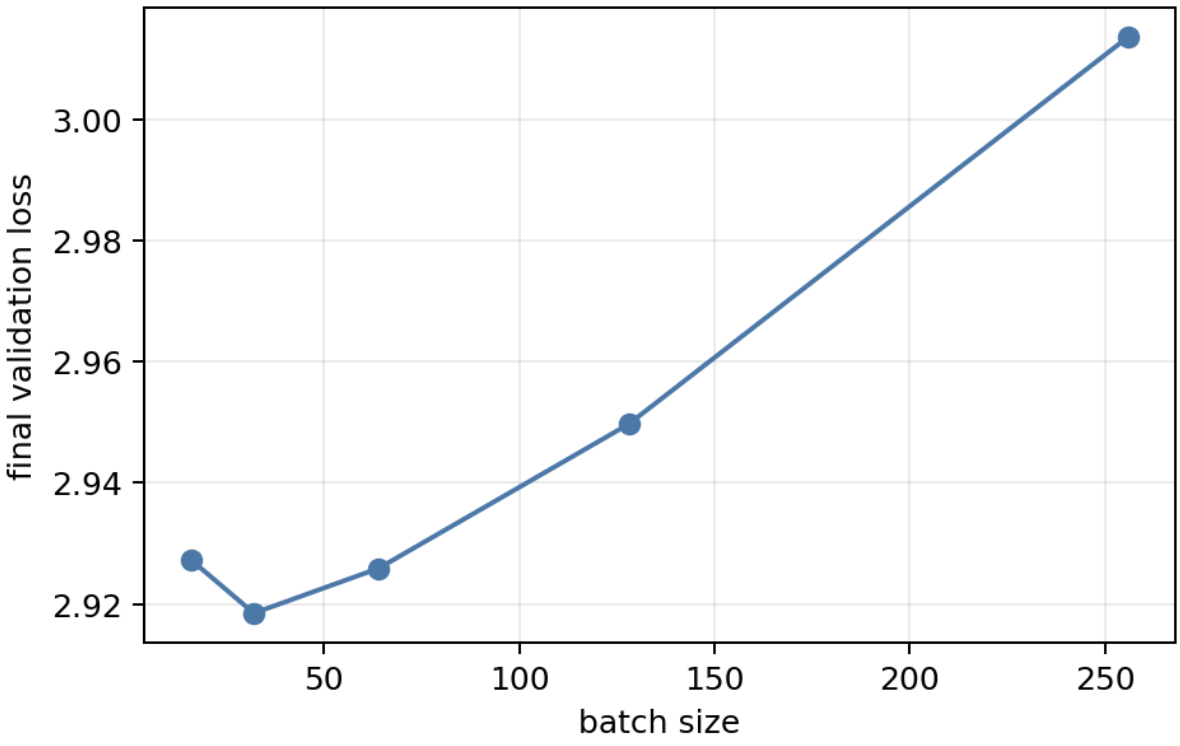


Figure 3: Batch-size sweep

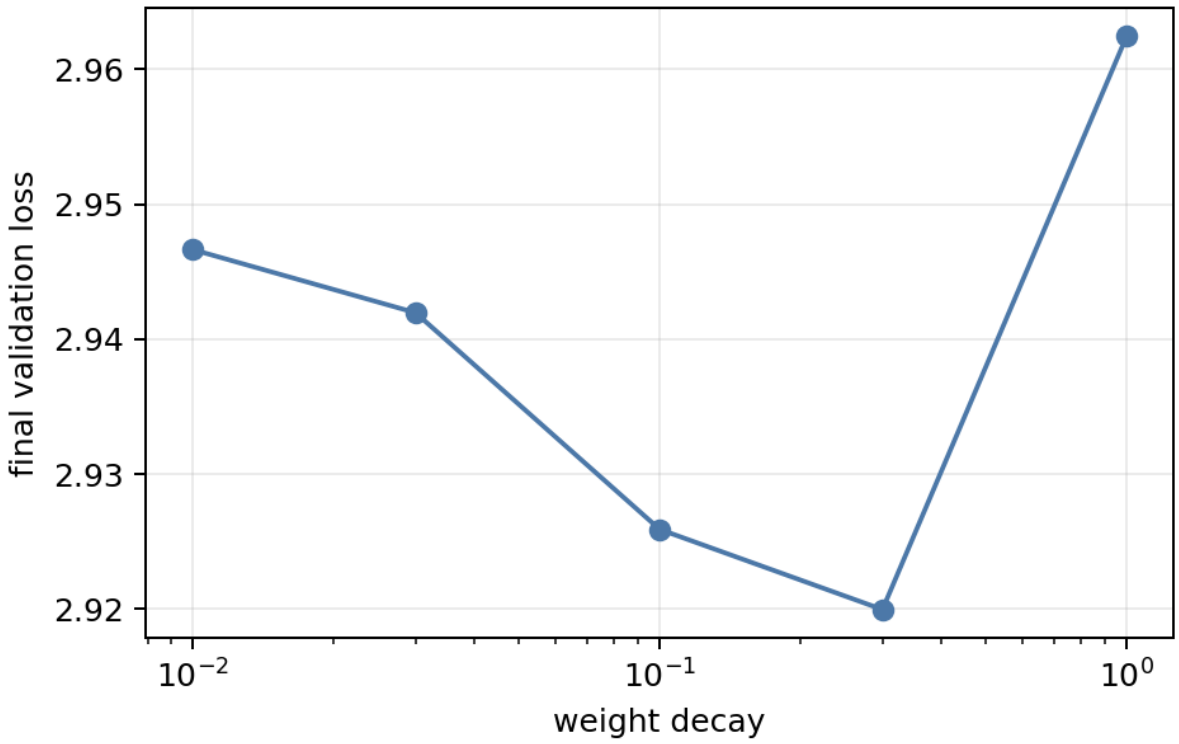


Figure 4: Weight-decay sweep

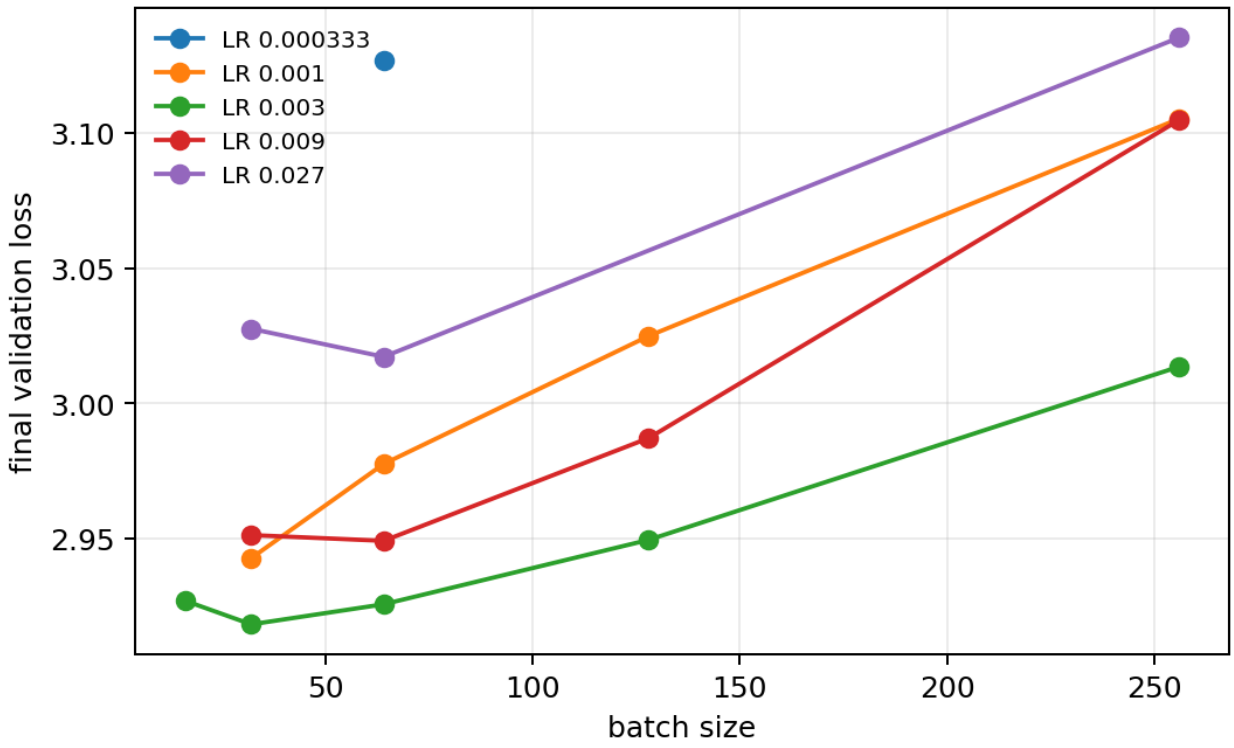


Figure 5: Learning rate and batch size

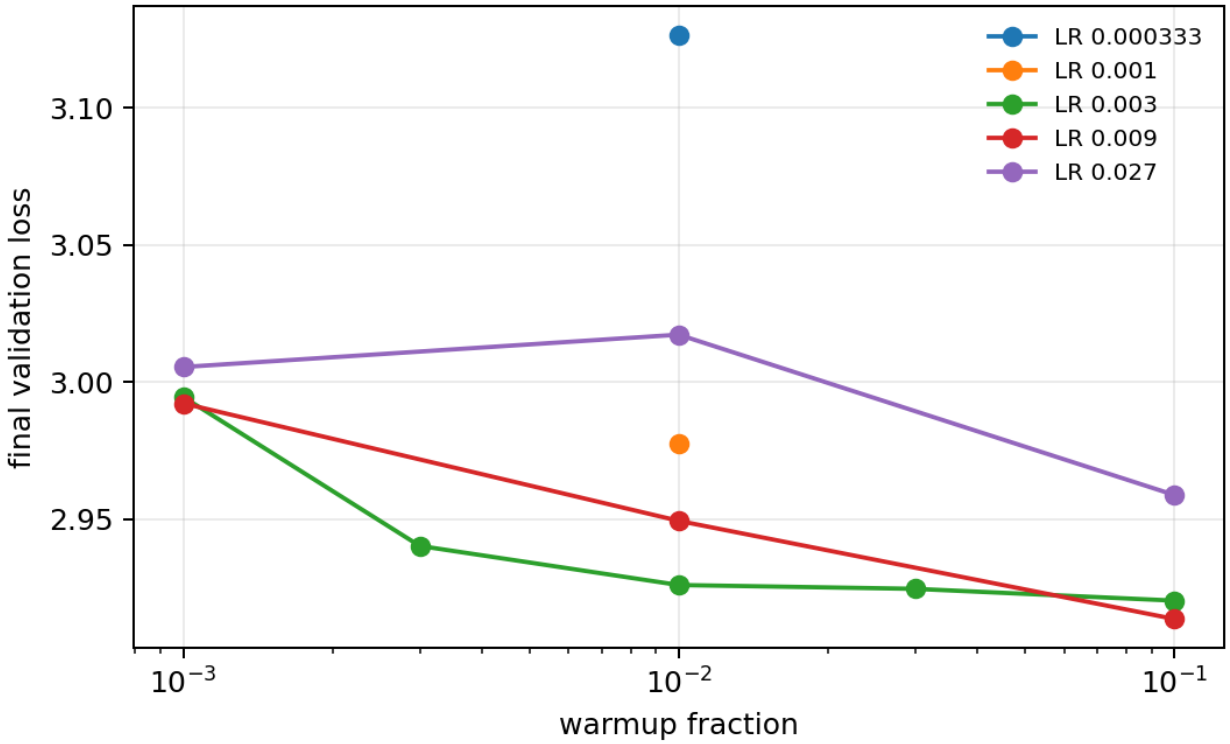


Figure 6: Learning rate and warmup

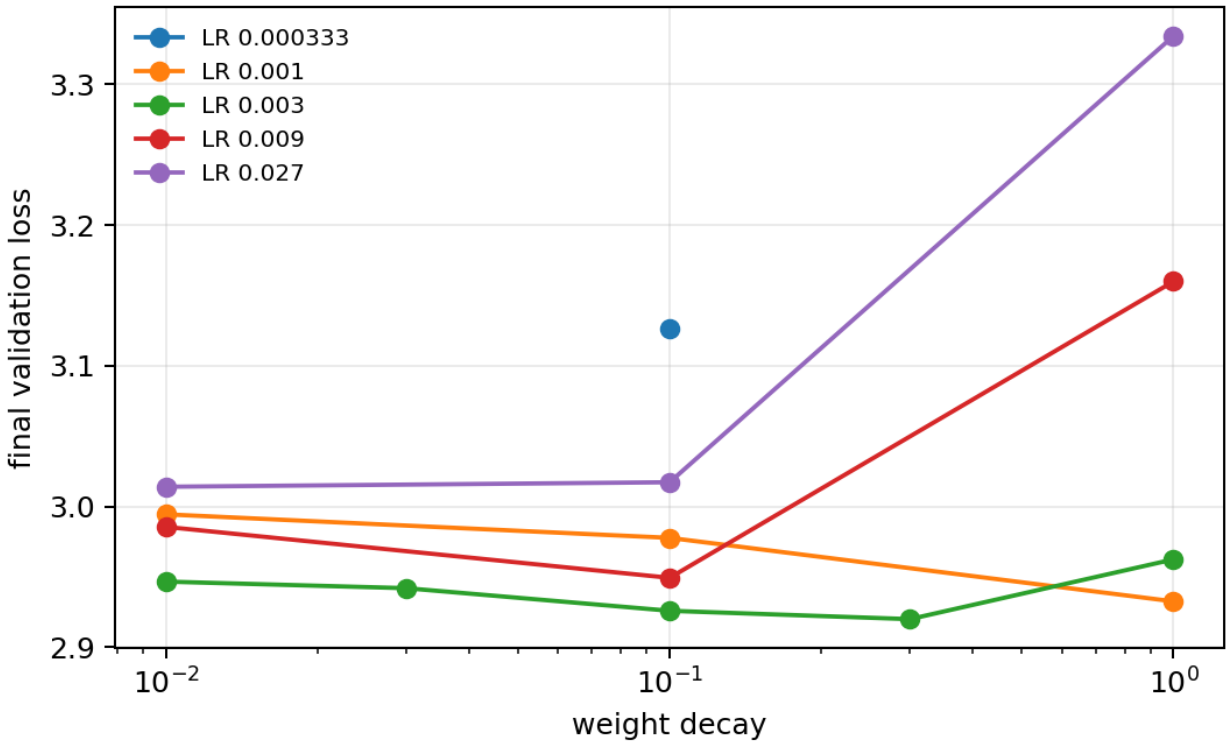


Figure 7: Learning rate and weight decay

(c) Learning-rate schedules

The schedule comparison crosses linear, cosine, constant, and warmup-stable-decay schedules with learning rate. Separate runs vary weight decay, the decay fraction, and warmup.

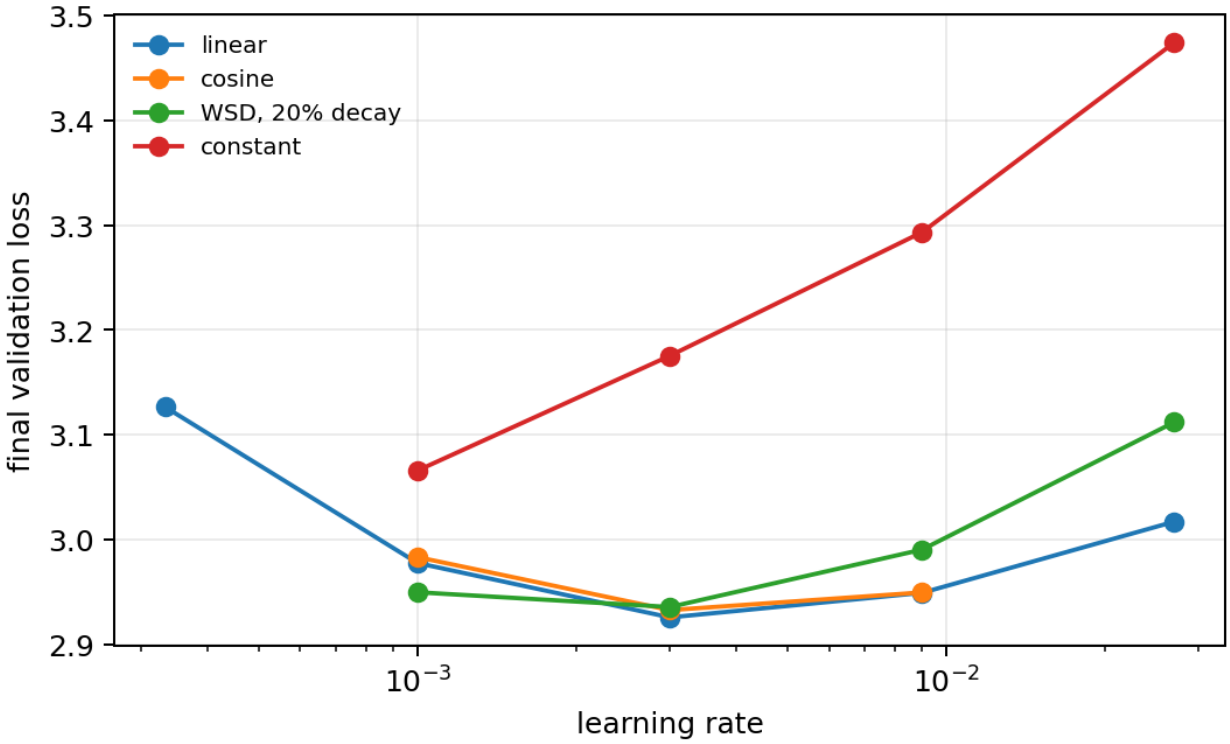


Figure 8: Learning rate and schedule

At learning rate 0.003, linear, cosine, warmup-stable-decay over the final 20%, and constant schedules finish at 2.9259, 2.9328, 2.9358, and 3.1754. Constant learning rate is worst at every tested learning rate because it never anneals. Cosine and warmup-stable-decay tolerate nearby learning rates, but neither improves on the standard linear schedule at its optimum.

Linear decay also remains best when weight decay is increased. At weight decay 1.0, linear, 20% warmup-stable-decay, and constant schedules give 2.9624, 3.0593, and 3.4861.

For warmup-stable-decay, allocating more training to the decay phase helps: decay fractions 0.1, 0.2, and 0.4 give 2.9554, 2.9358, and 2.9216. In the sample ranking experiment with learning rate 0.009 and weight decay 1.0, the linear schedule with 10% warmup is best (3.0566), followed by 20% warmup-stable-decay with 20% warmup (3.2082), then the same schedule without warmup (3.4174).

Synthesis

For this fixed d8 training budget, learning rate and batch size matter most. Learning rate 0.003 is the robust one-dimensional optimum, while batch size 32 is slightly better than 64 and much better than 128–256. Warmup and weight decay are forgiving near their defaults, but both interact with learning rate: 10% warmup makes learning rate 0.009 competitive, whereas weight decay 1.0 requires a lower learning rate. Annealing is essential; a constant schedule is consistently poor. The best observed losses, 2.9135–2.9216, come from modest coordinated changes rather than extreme values.

Problem 2: Reliability of scaling laws

The main scaling ladder uses depths 4–9 with width (64d), 614.4M fresh training tokens, and otherwise standard settings. Compute is approximated by (6ND), with non-embedding parameters N about 14.5 d w squared.

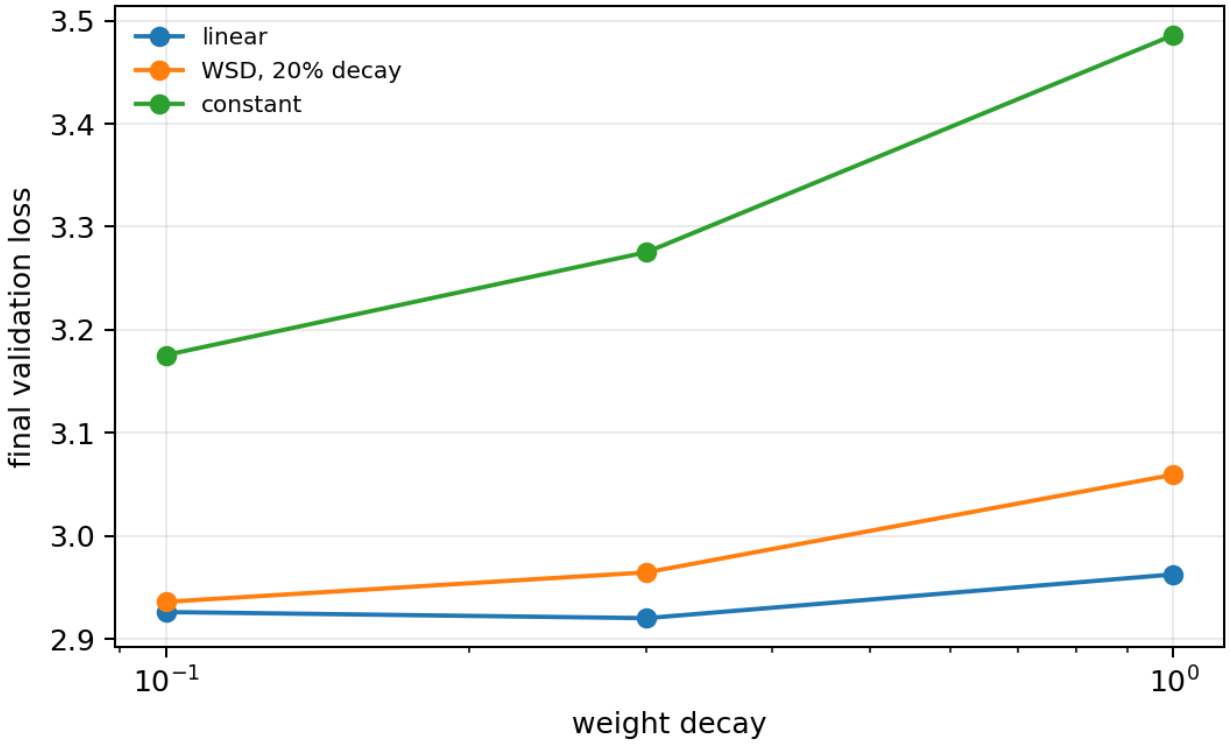


Figure 9: Weight decay and schedule

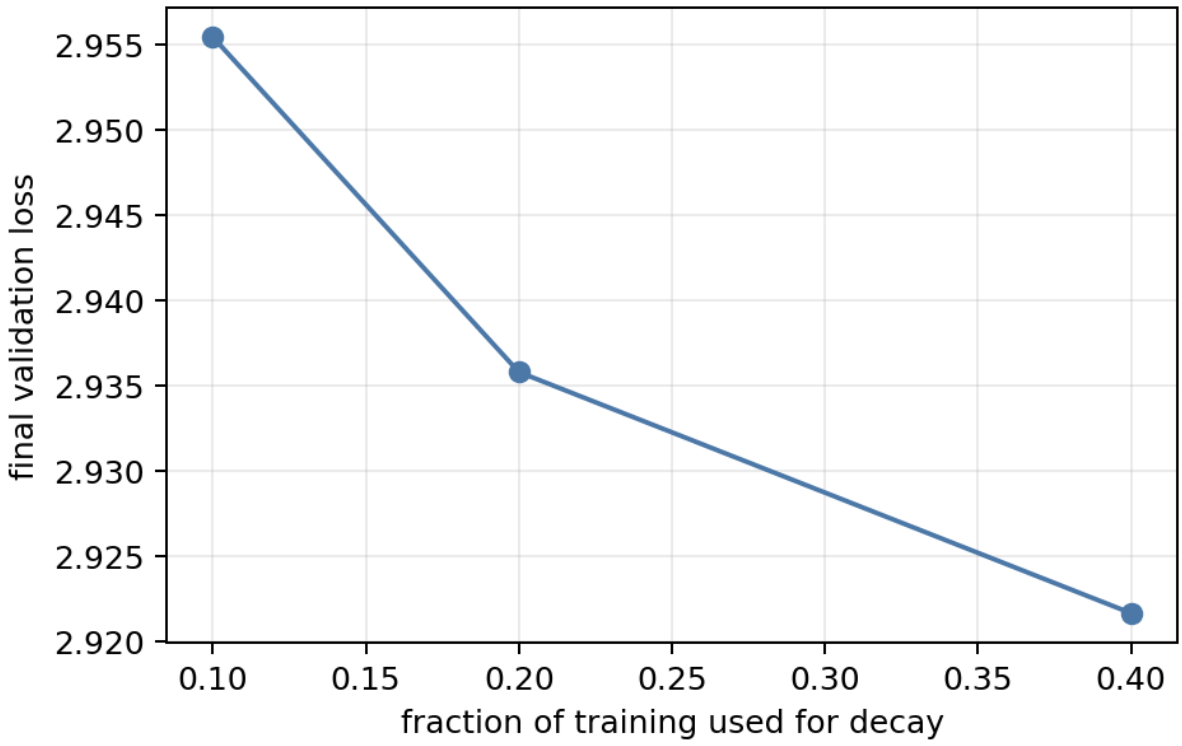


Figure 10: Warmup-stable-decay fraction

(a) Forecasting along a model-size ladder

Four depth ladders compare the standard recipe, a constant learning rate, dropout 0.2, and learning rate 0.03.

Depth	Standard	Constant schedule	Dropout 0.2	Learning rate 0.03
d4	3.2910	3.4256	3.4465	3.3178
d5	3.1538	3.3315	3.3257	3.2011
d6	3.0562	3.2472	3.2163	3.1096
d7	2.9854	3.2132	3.1294	3.0571
d8	2.9259	3.1754	3.0630	3.0307
d9	2.8837	3.1280	3.0033	3.0009

Fits using only d4–d7 were recorded before running d8 and d9:

Recipe	Predicted d8	Measured d8	Predicted d9	Measured d9	Predicted d20
Standard	2.929	2.926	2.884	2.884	2.664
Constant schedule	3.174	3.175	3.147	3.128	3.032
Dropout 0.2	3.053	3.063	2.986	3.003	2.530
Learning rate 0.03	3.009	3.031	2.974	3.001	2.810

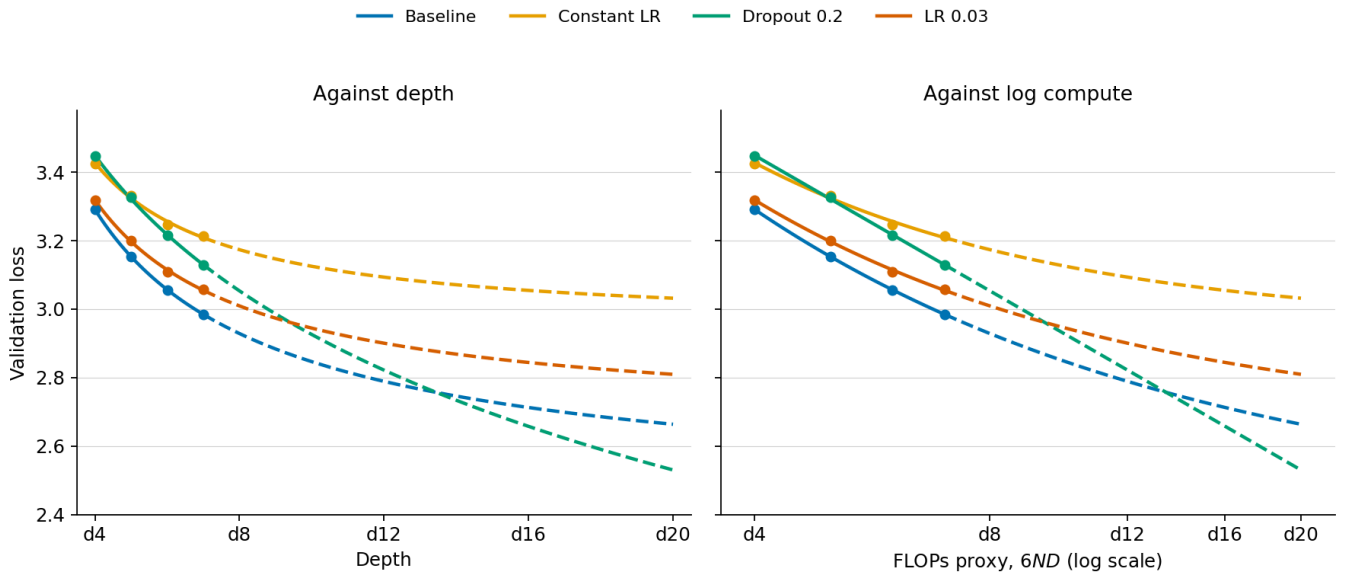


Figure 11: Fits from d4–d7 and extrapolations to d20

The standard and constant-schedule forecasts are accurate at d8 and d9. Dropout and the high learning rate are systematically worse than forecast by 0.01–0.03, suggesting curvature already visible at the held-out scales. The dropout line gives the lowest numerical d20 prediction and crosses the standard fit near d14, but this is the least trustworthy forecast because its d4–d7 fit has no identified loss floor. The standard d20 forecast of about 2.66 is the safer extrapolation.

(b) Changing the scaling axis or schedule

The schedule comparison holds data fixed and varies model size. The data comparison holds d8 fixed and doubles the number of fresh sequences from 75,000 to 2.4M.

Depth	Linear schedule	Cosine schedule
d4	3.2910	3.2939
d5	3.1538	3.1574
d6	3.0562	3.0613
d7	2.9854	2.9904
d8	2.9259	2.9328
d9	2.8837	2.8917

Training tokens	Final validation loss
76.8M	3.5344
153.6M	3.2307
307.2M	3.0584
614.4M	2.9259
1.23B	2.8371
2.46B	2.7682

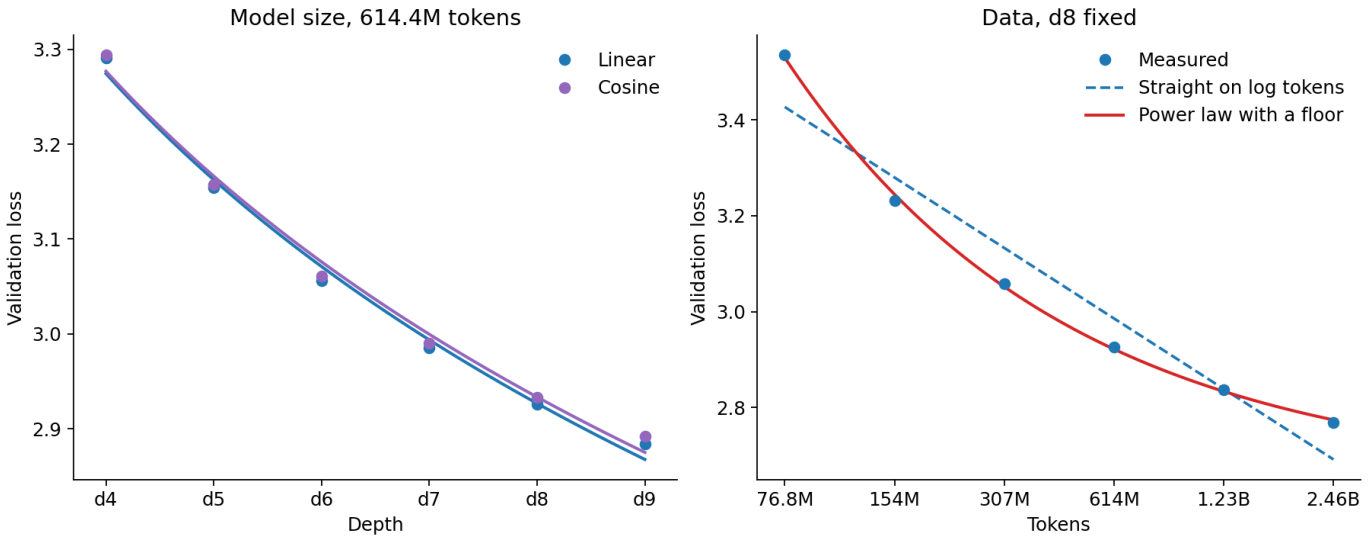


Figure 12: Model-size and data-scaling curves

Linear and cosine schedules have nearly identical model-size slopes: loss falls by 0.116 and 0.115 per doubling of the compute proxy, respectively, with R-squared about 0.99. Cosine is slightly worse at every depth, so it changes the intercept rather than the slope.

Data scaling is steeper but visibly curved. Successive token doublings improve validation loss by 0.304, 0.172, 0.133, 0.089, and 0.069. A straight line in log tokens reaches only an R-squared of about 0.93, while a power law with a floor near 2.65 fits all six points within about 0.01. Thus the measured data slope depends strongly on the token range.

(c) Breaking the trend

Two interventions test data limitation: repeatedly training on the same 75,000 sequences and scaling model depth at a fixed budget of 76.8M fresh tokens. Two controls vary width at depth 8 and depth at width 512 while retaining 614.4M fresh tokens.

Epochs over 75,000 sequences	Tokens processed	Repeated data	Fresh data
1	76.8M	3.5344	3.5344
2	153.6M	3.2431	3.2307

Epochs over 75,000 sequences	Tokens processed	Repeated data	Fresh data
4	307.2M	3.1121	3.0584
8	614.4M	**3.0896**	2.9259
16	1.23B	3.1607	2.8371
32	2.46B	3.2504	2.7682

Depth	76.8M tokens	614.4M tokens
d4	3.7638	3.2910
d5	3.6277	3.1538
d6	3.5756	3.0562
d7	3.5871	2.9854
d8	3.5344	2.9259
d9	3.5838	2.8837

Repeated data follows fresh-data performance for roughly two epochs, then overfitting becomes visible. Validation loss bottoms out at eight epochs and rises thereafter, while fresh-data runs continue improving. Similarly, the 76.8M-token depth ladder improves only through d6 and then fluctuates; its log-compute fit has slope 0.052 and R-squared about 0.71, compared with slope 0.116 and R-squared about 0.99 at 614.4M tokens.

Changing model shape does not by itself break the trend when enough fresh data is available. At fixed width 512, depths 4, 5, 6, 7, 8, 9, and 12 give losses 3.0556, 3.0157, 2.9783, 2.9515, 2.9259, 2.9065, and 2.8733. At fixed depth 8, widths 256, 320, 384, 448, 512, 576, and 768 give 3.1473, 3.0719, 3.0176, 2.9663, 2.9259, 2.8993, and 2.8455. Both remain smooth and approximately linear in log compute, although the widest model gains less than the joint depth-width ladder predicts.

Synthesis

Scaling is highly predictable only within a well-supported regime. The standard d4–d7 fit predicts d8 and d9 almost exactly, and changing linear decay to cosine preserves the slope. The reliability fails when scale increases capacity without enough distinct data: repeating a small dataset eventually worsens validation loss, and a depth ladder with only 76.8M tokens becomes non-monotonic beyond d6. Fresh data restores smooth scaling, even when depth and width are varied separately. A good fit is therefore evidence about a particular model-data recipe, not a universal law.

Problem 3: Measuring variation

All comparisons use the standard d8 recipe unless an intervention is named. Initialization is controlled by the model seed, data order by the data seed, and fixed-device reproducibility by deterministic execution.

(a) Joint sources of variation

Three runs vary model and data seeds together. Two use H100s and one uses an A100, so the comparison includes initialization, data order, and hardware.

Model/data seed	Hardware	Final validation loss
1	H100	2.9285
2	H100	2.9303
3	A100	2.9328

The mean is 2.9305, the sample standard deviation is 0.0022, and the range is 0.0044.

The curves are visually indistinguishable at full scale. Three points are too few to establish a distributional shape, but they place ordinary terminal-loss variation at a few thousandths of a nat.

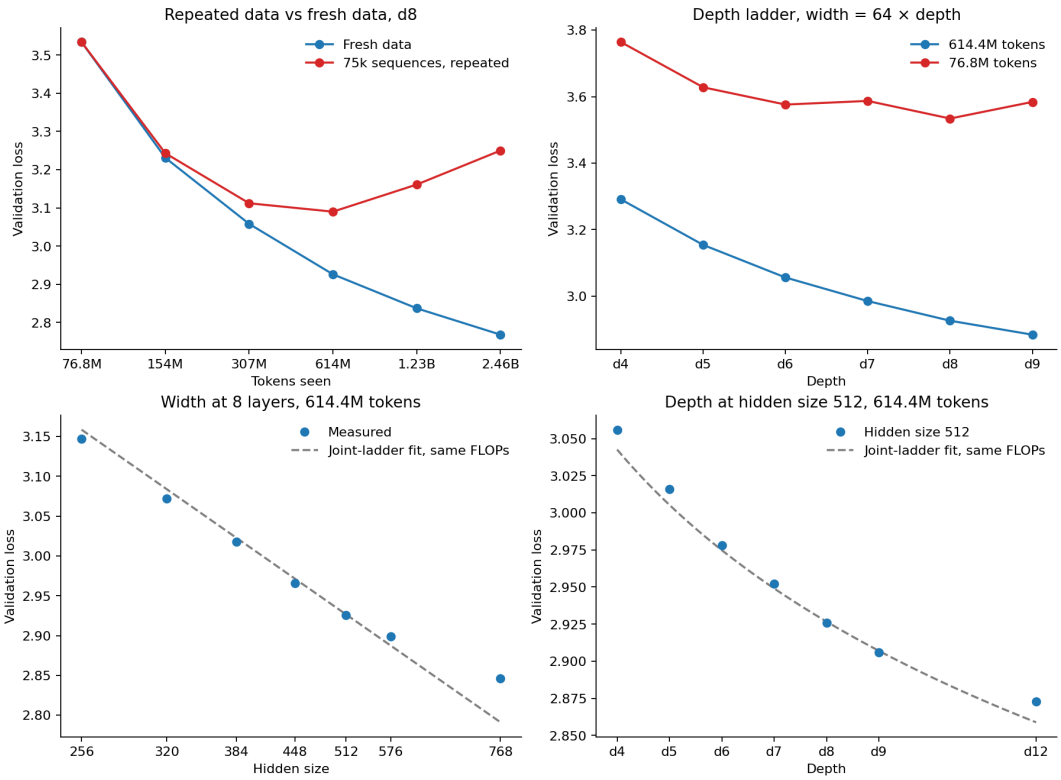


Figure 13: Interventions that preserve or break scaling

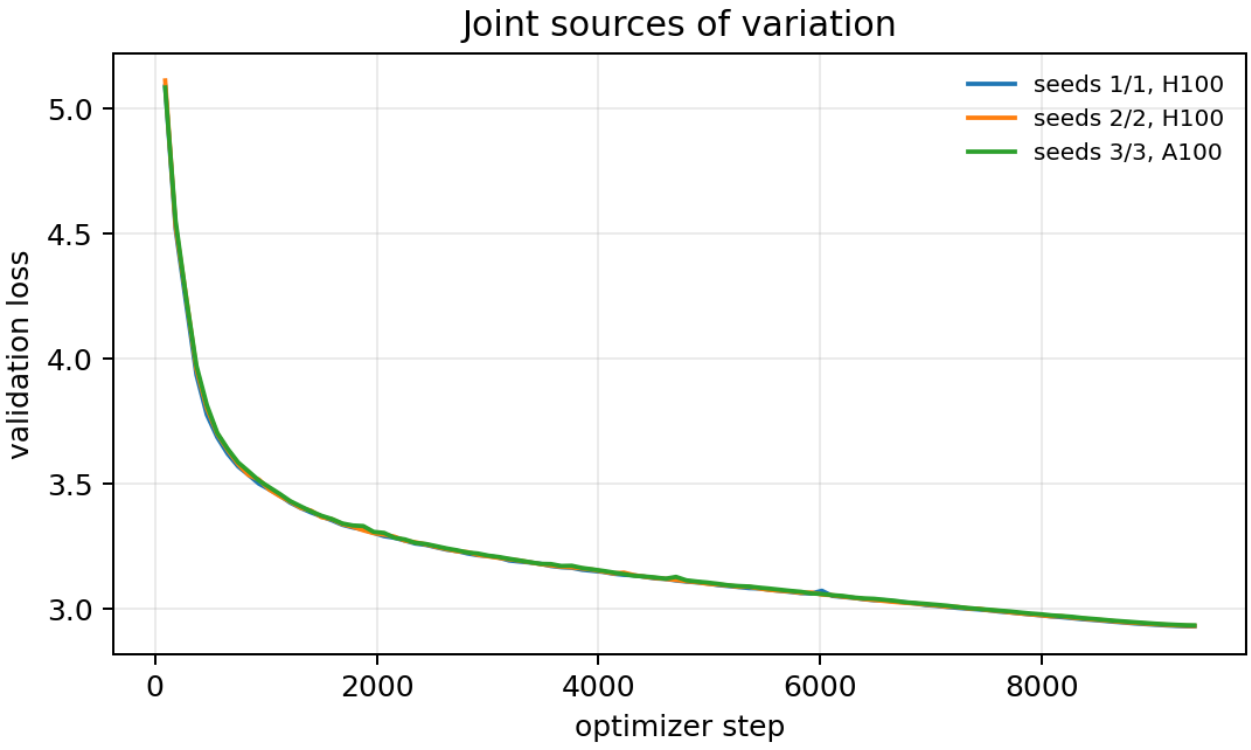


Figure 14: Validation curves with all sources varied

(b) Isolating each source

Initialization and data order are varied separately under deterministic execution on H100s. Two identical deterministic runs test exact replay.

Source varied	Final validation losses	Sample SD	Range
Initialization	2.9298, 2.9244, 2.9301	0.0032	0.0057
Data order	2.9309, 2.9272, 2.9286	0.0019	0.0037
Deterministic replay	2.9271, 2.9271	0	0
Joint sources from part (a)	2.9285, 2.9303, 2.9328	0.0022	0.0044

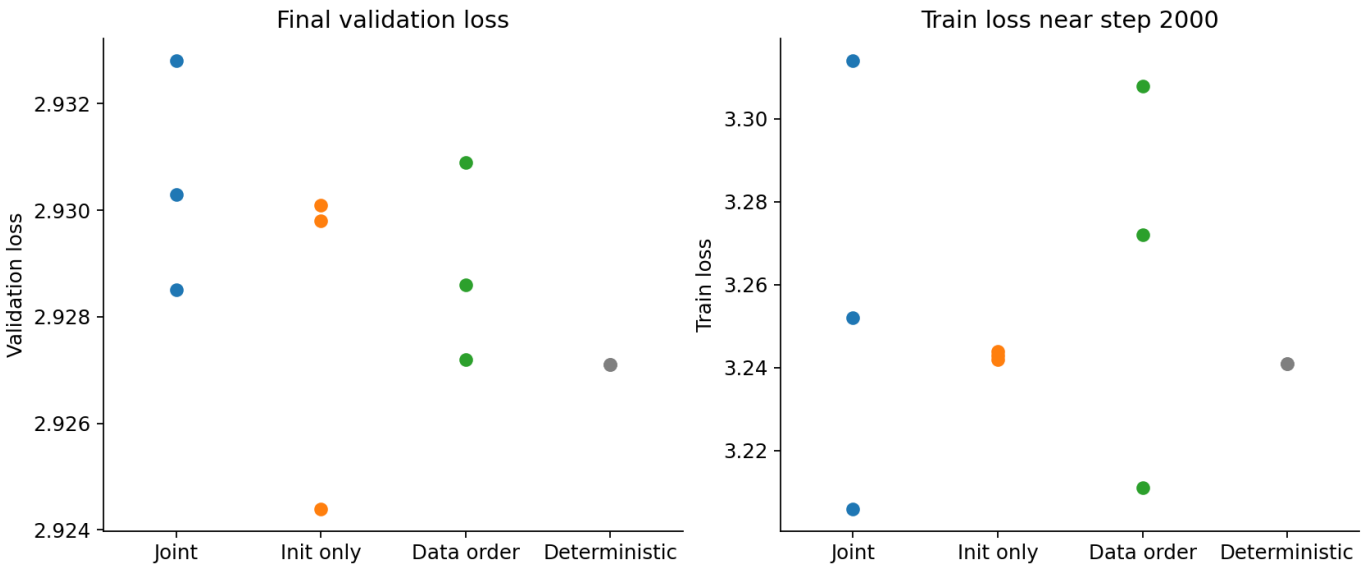


Figure 15: Terminal validation loss and mid-training train loss

Initialization produces the largest terminal spread. Data order has a smaller final effect but a much larger effect on individual train batches: near step 2000, data-order runs span train losses 3.211–3.308, while initialization runs span only 3.242–3.244. This occurs because changing data order changes the batch being measured as well as the optimization path.

The two deterministic replays match exactly to the recorded precision, showing that the remaining spread is attributable to controlled seeds rather than irreproducibility on a fixed device. The A100 joint run lies within the same narrow band. A dedicated same-seed A100 run ended early, so hardware was not cleanly isolated; the cross-hardware point is only evidence that any hardware shift is not large at this resolution.

(c) Hyperparameters and scale

Seeds 4–8 finished for the standard recipe, batch size 16, and learning rate 0.009. Dropout, learning rate 0.027, d12, and the 2.46B-token run finished for seeds 0–2.

Setting	Seeds	Mean validation loss	Sample SD	Range
Standard d8	4–8	2.9329	0.0025	0.0064
Batch size 16	4–8	2.9352	0.0039	0.0095
Learning rate 0.009	4–8	2.9489	0.0058	0.0142
Dropout 0.2	0–2	3.0600	0.0044	0.0086
Learning rate 0.027	0–2	3.0047	0.0058	0.0105
d12	0–2	2.7910	0.0049	0.0087
d8, 2.46B tokens	0–2	2.7705	0.0024	0.0048

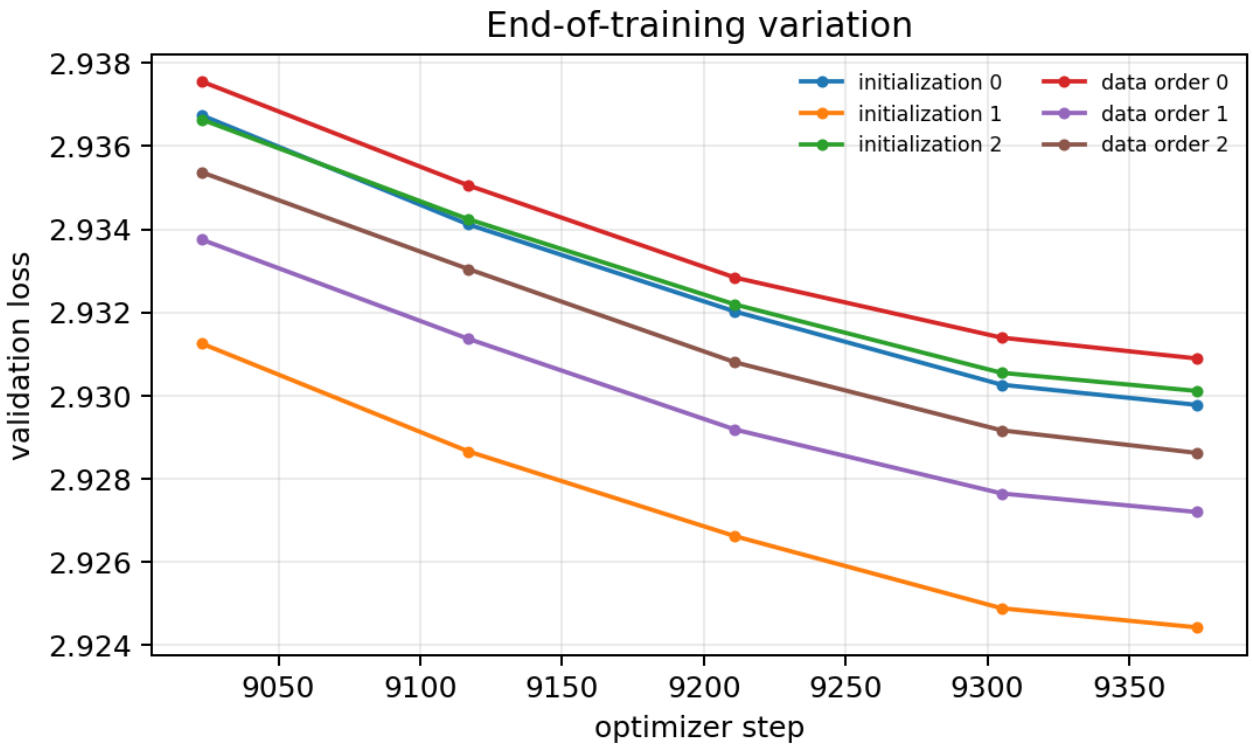


Figure 16: End-of-training validation curves

Learning rate 0.009 has 2.3 times the standard deviation of the standard recipe. Batch size 16 has 1.6 times. Only the higher learning rate clears a twofold increase. Dropout 0.2 and learning rate 0.027 are also wider than the standard run. The d12 mean is lower (2.791 versus 2.933) and its standard deviation is 0.0049, so the lower loss did not shrink the spread. Four times more tokens lowers the mean to 2.770 and leaves the spread unchanged at 0.0024.

Synthesis

On the standard d8 recipe, terminal validation noise is a few thousandths of a nat. Initialization contributes more of that spread than data order. Data order moves individual train batches more, because it changes which batch is logged. Deterministic replay on one H100 matches exactly. A gap near 0.001 is inside this noise. A repeatable gap near 0.01 is not. A higher learning rate widens the distribution, so the noise floor has to be measured for the recipe being compared. Training four times longer lowers the loss and does not shrink that floor.

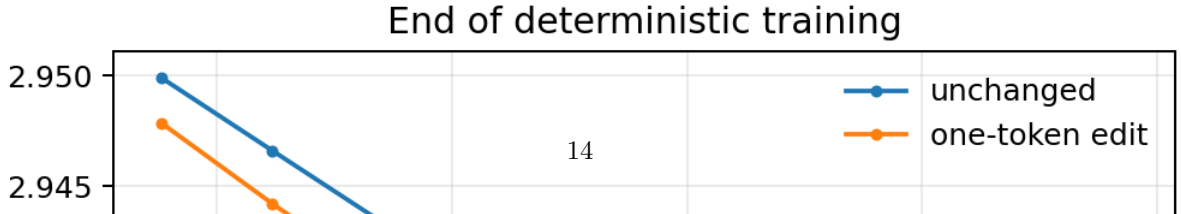
Problem 4: Amplification of randomness

All ordinary randomness is fixed with deterministic execution. The standard d8 recipe is used throughout.

(a) A one-token change

The perturbed run changes token position 100 in the first shuffled training sequence to token ID 17. Every other input, seed, and training setting is identical.

Training data	Final train loss	Final validation loss
Unchanged	2.8859	2.9277
One-token edit	2.8858	2.9254



ordinary seed spread measured in Problem 3. The edited run happens to be lower; this is not evidence of a real improvement.

(b) Perturbation time and magnitude

At optimizer step (T), independent Gaussian noise is added to every parameter tensor. The noise standard deviation is the stated scale times that tensor's own standard deviation, after which training continues to step 9375. Four perturbation times and three magnitudes test whether the optimizer forgets early or small changes.

Perturbation step T	Updates remaining	scale 0.001	scale 0.01	scale 0.1
100	9275	2.9258	2.9261	2.9259
2000	7375	2.9262	2.9264	2.9265
5000	4375	2.9262	2.9258	2.9268
8000	1375	2.9262	2.9262	2.9301

The unperturbed run ended at 2.9277.

Eleven of the twelve kicks finish between 2.9258 and 2.9268. Time and magnitude do not matter when thousands of steps remain, including a scale of 0.1 at step 100. The exception is scale 0.1 at step 8000, with 1,375 steps left: it finishes at 2.9301, about 0.004 above the two smaller kicks at that same step. A large late kick is the one that is still visible in the terminal loss.

Synthesis

A one-token edit shifts the validation curve by about 0.002, inside the seed spread from Problem 3. A weight kick is forgotten by the end unless it is large and late. Scale 0.1 at step 8000 leaves a 0.004 gap. Every earlier kick, and the smaller kicks at step 8000, land on the same validation loss.

Problem 5: Loss-curve augury

These comparisons reuse d8 runs trained on 614.4M tokens. The standard recipe finishes at validation loss 2.9259.

(a) Macro shape

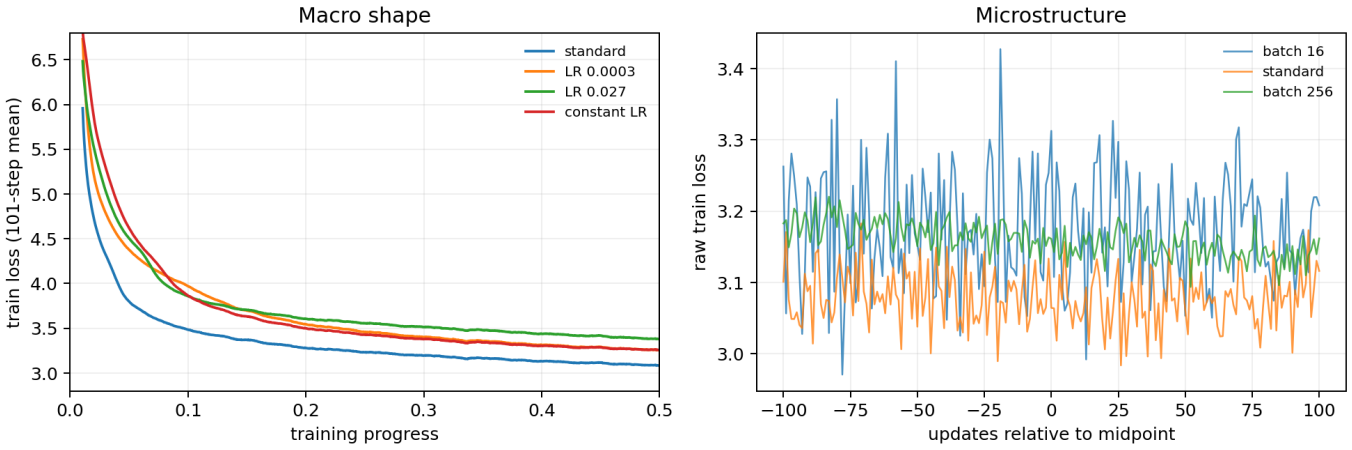


Figure 18: Macro loss-curve shape and batch-dependent microstructure

Healthy runs share a steep initial descent, a long region of diminishing returns, and a final improvement as the learning rate anneals. Learning rate and schedule control this large-scale shape:

- At 10%, 50%, and 90% of training, the standard run's local mean train losses are 3.466, 3.075, and 2.929.

- Learning rate 0.0003 descends too slowly, with corresponding losses 3.932, 3.247, and 3.136 and final validation loss 3.1473.
- Learning rate 0.027 also starts slowly and catches up late, reaching 3.833, 3.372, and 3.094 before finishing at validation loss 3.0172.
- Constant learning rate reaches 3.825, 3.248, and 3.167 and then remains high, finishing at 3.1754. Without annealing, it misses the standard run's late improvement.
- Cosine decay is smooth throughout. Warmup-stable-decay stays flatter through its stable phase and improves sharply only when decay begins.

Longer warmup delays the early descent but can help a high learning rate avoid a poor trajectory. Weight decay mostly changes the attained loss rather than producing a distinctive curve shape.

(b) Microstructure

Batch size controls short-timescale noise. Measured as the standard deviation of adjacent train-loss differences between 10% and 90% of training, batch sizes 16, 32, 64, 128, and 256 give 0.111, 0.080, 0.056, 0.039, and 0.028. This is close to the expected inverse-square-root dependence on batch size.

Changing learning rate from 0.0003 to 0.027 at fixed batch size leaves this statistic near 0.055. Schedules also remain near 0.055. Learning rate and schedule therefore alter the trend much more than the local jaggedness; batch size is the dominant smoothness control.

(c) Diagnostic patterns

Good d8 training is finite, steadily improving after warmup, and benefits from late annealing. Several recognizable failure modes appear in the completed runs:

- A learning rate that is too small produces a smooth but uniformly slow descent.
- A learning rate that is too large produces a delayed, high-loss trajectory that may recover late but finishes worse.
- A constant schedule appears reasonable early, then plateaus above an annealed run.
- A very large batch is visually smooth but learns less effectively at the fixed token budget.
- Warmup-stable-decay produces a deliberate late bend rather than a failure.

For the sample identification task, the high-loss early curve is most consistent with learning rate 0.009, not the standard, low-learning-rate, or batch-256 run. Its final validation loss is 2.9493, only 0.0234 above the standard run, so it belongs in the "within +0.05" bucket.

Synthesis

Macro shape and microstructure diagnose different controls. Learning rate, warmup, and schedule determine when useful learning occurs and whether the run receives a late annealing gain. Batch size primarily determines point-to-point noise: increasing it makes the curve smoother but does not guarantee a better model. On d8, a smooth curve can still be undertrained, while a noisy small-batch curve can finish well. The most reliable healthy pattern is rapid early progress followed by continued slow improvement and a final benefit from decay.

Problem 6: Activations and gradient norms

RMS on one training batch, before clipping. The standard run is the reference.

(a) Standard d8

Global RMS	Start	Middle	End
Parameters	0.0422	0.0978	0.0892
Activations	0.937	3.157	2.593
Gradients	(2.68×10^{-3})	(1.85×10^{-5})	(2.25×10^{-5})

Parameters and activations peak mid-run, then fall as the learning rate decays. Gradients drop about 100× after step 0 and stay near (2×10^{-5}).

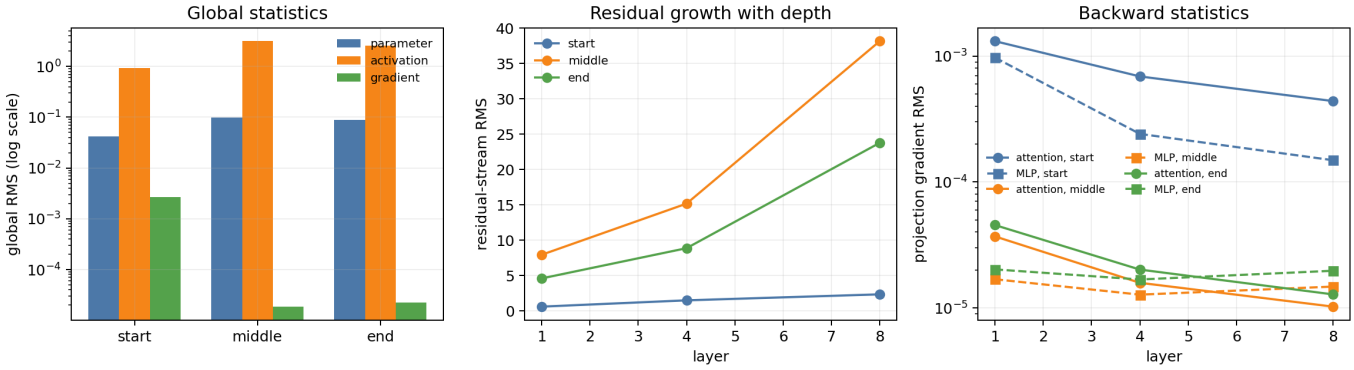


Figure 19: Parameter, activation, and gradient RMS through training and depth

Residual RMS grows with depth. Layers 1, 4, and 8 end at 4.58, 8.86, and 23.8. At layer 8 the MLP output ends larger than attention (8.67 vs 4.28). At step 0, attention-output gradient RMS is (1.32×10^{-3}) in layer 1 and (4.39×10^{-4}) in layer 8.

(b) Earlier interventions

Terminal global RMS.

Intervention	Parameter RMS	Activation RMS	Gradient RMS
Standard	0.0892	2.593	(2.25×10^{-5})
Learning rate 0.001	0.0522	1.844	(3.72×10^{-5})
Learning rate 0.009	0.1621	3.849	(1.67×10^{-5})
Learning rate 0.027	0.3659	7.909	(1.31×10^{-5})
Batch size 16	0.1012	2.352	(6.59×10^{-5})
Batch size 256	0.0670	2.292	(1.46×10^{-5})
Weight decay 1.0	0.0494	1.283	(8.16×10^{-5})
Constant learning rate	0.1319	4.773	(1.69×10^{-5})
Dropout 0.2	0.0894	2.422	(1.91×10^{-5})

Raising learning rate from 0.001 to 0.027 multiplies terminal parameter RMS by 7.0 and activation RMS by 4.3. Weight decay 1.0 roughly halves both. Batch 16 has 4.5× the terminal gradient RMS of batch 256. Dropout barely moves the globals. Constant learning rate stays high at the end (activations 4.77); cosine stays near the default (2.74). Warmup 0.10 does not change the end (activations 2.62).

(c) Moving the statistics

- Raise the learning rate to enlarge parameters and activations for the whole run.
- Raise weight decay, or lower the learning rate, to shrink them.
- Remove learning-rate decay to keep them large at the end. Warmup only moves the start.
- Shrink the batch to raise terminal gradient RMS.

Synthesis

Activations accumulate with depth and peak mid-training. Gradients are large only at the start. Learning rate and weight decay set the scale; batch size sets the gradient; the schedule decides whether the scale lasts to the end.

Problem 7: Optional prediction problem

No optional prediction problem is included in this submission.

Assignment 1 run index

Completed runs with a recorded final validation loss. Interrupted runs are omitted. A run reused later is listed once.

Unless a row says otherwise, the configuration is d8 (8 layers, width 512), 614.4M fresh tokens, AdamW, learning rate 0.003, 1% warmup, linear decay, batch size 64, weight decay 0.1, no dropout, and model/data seed 42.

Problem 1(a): one-variable sweeps

ID	Changes from the standard recipe	Final train loss	Final validation loss
1A-01	learning rate 0.001	2.9360	2.9777
1A-02	learning rate 0.000333	3.0909	3.1265
1A-03	warmup 0.1%	2.9507	2.9945
1A-04	warmup 0.3%	2.8995	2.9402
1A-05	standard recipe	2.8859	2.9259
1A-06	learning rate 0.009	2.9049	2.9493
1A-07	learning rate 0.027	2.9735	3.0172
1A-08	warmup 3.0%	2.8815	2.9245
1A-09	warmup 10.0%	2.8788	2.9202
1A-10	batch 16	2.7977	2.9272
1A-11	batch 32	2.8631	2.9185
1A-12	batch 128	2.9087	2.9497
1A-13	weight decay 0.01	2.9036	2.9466
1A-14	weight decay 0.03	2.9051	2.9420
1A-15	weight decay 0.3	2.8761	2.9200
1A-16	weight decay 1	2.9160	2.9624
1A-17	batch 256	3.0050	3.0136

Problem 1(b): hyperparameter interactions

ID	Changes from the standard recipe	Final train loss	Final validation loss
1B-01	learning rate 0.001; batch 256	3.0983	3.1051
1B-02	learning rate 0.009; batch 256	3.0967	3.1046
1B-03	learning rate 0.027; batch 256	3.1269	3.1350
1B-04	learning rate 0.001; batch 32	2.8895	2.9429
1B-05	learning rate 0.009; batch 32	2.8989	2.9514
1B-06	learning rate 0.027; batch 32	2.9739	3.0276
1B-07	learning rate 0.009; warmup 0.1%	2.9542	2.9922
1B-08	learning rate 0.009; warmup 10.0%	2.8670	2.9135
1B-09	learning rate 0.027; warmup 0.1%	2.9625	3.0054
1B-10	learning rate 0.027; warmup 10.0%	2.9122	2.9588
1B-11	learning rate 0.001; weight decay 0.01	2.9526	2.9944

ID	Changes from the standard recipe	Final train loss	Final validation loss
1B-12	learning rate 0.001; weight decay 1	2.8873	2.9326
1B-13	learning rate 0.009; weight decay 0.01	2.9413	2.9855
1B-14	learning rate 0.009; weight decay 1	3.1170	3.1599
1B-15	learning rate 0.027; weight decay 0.01	2.9682	3.0140
1B-16	learning rate 0.027; weight decay 1	3.2966	3.3343
1B-17	learning rate 0.001; batch 128	2.9835	3.0249
1B-18	learning rate 0.009; batch 128	2.9429	2.9873

Problem 1(c): learning-rate schedules

ID	Changes from the standard recipe	Final train loss	Final validation loss
1C-01	learning rate 0.001; cosine schedule	2.9410	2.9833
1C-02	cosine schedule	2.8909	2.9328
1C-03	learning rate 0.001; constant schedule	3.0316	3.0659
1C-04	learning rate 0.009; cosine schedule	2.9067	2.9497
1C-05	constant schedule	3.1424	3.1754
1C-06	warmup-stable-decay (20% decay)	2.8926	2.9358
1C-07	learning rate 0.001; warmup-stable-decay (20% decay)	2.9051	2.9499
1C-08	learning rate 0.009; constant schedule	3.2627	3.2932
1C-09	learning rate 0.009; warmup-stable-decay (20% decay)	2.9470	2.9906
1C-10	learning rate 0.027; constant schedule	3.4480	3.4741
1C-11	learning rate 0.027; warmup-stable-decay (20% decay)	3.0653	3.1121
1C-12	constant schedule; weight decay 0.3	3.2375	3.2753
1C-13	warmup-stable-decay (20% decay); weight decay 0.3	2.9192	2.9644
1C-14	constant schedule; weight decay 1	3.4484	3.4861
1C-15	warmup-stable-decay (20% decay); weight decay 1	3.0113	3.0593

ID	Changes from the standard recipe	Final train loss	Final validation loss
1C-16	warmup-stable-decay (10% decay)	2.9081	2.9554
1C-17	warmup-stable-decay (40% decay)	2.8747	2.9216
1C-18	learning rate 0.009; warmup-stable-decay (20% decay); warmup 0.0%; weight decay 1	3.3788	3.4174
1C-19	learning rate 0.009; warmup-stable-decay (20% decay); warmup 20.0%; weight decay 1	3.1639	3.2082
1C-20	learning rate 0.009; warmup 10.0%; weight decay 1	3.0117	3.0566

Problem 2(a): model-size ladders

ID	Changes from the standard recipe	Final train loss	Final validation loss
2A-01	4 layers, width 256	3.2539	3.2910
2A-02	4 layers, width 256; dropout 0.2	3.5533	3.4465
2A-03	4 layers, width 256; learning rate 0.03	3.2766	3.3178
2A-04	4 layers, width 256; constant schedule	3.3908	3.4256
2A-05	5 layers, width 320; constant schedule	3.2950	3.3315
2A-06	5 layers, width 320; dropout 0.2	3.4096	3.3257
2A-07	5 layers, width 320	3.1110	3.1538
2A-08	5 layers, width 320; learning rate 0.03	3.1591	3.2011
2A-09	6 layers, width 384; constant schedule	3.2053	3.2472
2A-10	6 layers, width 384	3.0175	3.0562
2A-11	6 layers, width 384; learning rate 0.03	3.0660	3.1096
2A-12	6 layers, width 384; dropout 0.2	3.2908	3.2163
2A-13	7 layers, width 448	2.9417	2.9854
2A-14	7 layers, width 448; constant schedule	3.1751	3.2132
2A-15	7 layers, width 448; dropout 0.2	3.1910	3.1294
2A-16	7 layers, width 448; learning rate 0.03	3.0166	3.0571
2A-17	learning rate 0.03	2.9916	3.0307
2A-18	dropout 0.2	3.1125	3.0630
2A-19	9 layers, width 576	2.8436	2.8837
2A-20	9 layers, width 576; constant schedule	3.0930	3.1280

ID	Changes from the standard recipe	Final train loss	Final validation loss
2A-21	9 layers, width 576; dropout 0.2	3.0466	3.0033
2A-22	9 layers, width 576; learning rate 0.03	2.9574	3.0009

Problem 2(b): alternate scaling axes

ID	Changes from the standard recipe	Final train loss	Final validation loss
2B-01	75,000 sequences (76.8M tokens processed)	3.4948	3.5344
2B-02	4 layers, width 256; cosine schedule	3.2522	3.2939
2B-03	1,200,000 sequences (1.23B tokens processed)	2.8325	2.8371
2B-04	2,400,000 sequences (2.46B tokens processed)	2.7565	2.7682
2B-05	150,000 sequences (154M tokens processed)	3.1957	3.2307
2B-06	300,000 sequences (307M tokens processed)	3.0878	3.0584
2B-07	6 layers, width 384; cosine schedule	3.0183	3.0613
2B-08	5 layers, width 320; cosine schedule	3.1193	3.1574
2B-09	9 layers, width 576; cosine schedule	2.8537	2.8917
2B-10	7 layers, width 448; cosine schedule	2.9474	2.9904

Problem 2(c): scaling-law failures

ID	Changes from the standard recipe	Final train loss	Final validation loss
2C-01	75,000 sequences × 2 epochs (154M tokens processed)	3.1562	3.2431
2C-02	75,000 sequences × 16 epochs (1.23B tokens processed)	2.5468	3.1607
2C-03	75,000 sequences × 4 epochs (307M tokens processed)	2.9157	3.1121
2C-04	75,000 sequences × 8 epochs (614M tokens processed)	2.7004	3.0896

ID	Changes from the standard recipe	Final train loss	Final validation loss
2C-05	75,000 sequences × 32 epochs (2.46B tokens processed)	2.4084	3.2504
2C-06	4 layers, width 256; 75,000 sequences (76.8M tokens processed)	3.7230	3.7638
2C-07	5 layers, width 320; 75,000 sequences (76.8M tokens processed)	3.5964	3.6277
2C-08	6 layers, width 384; 75,000 sequences (76.8M tokens processed)	3.5460	3.5756
2C-09	7 layers, width 448; 75,000 sequences (76.8M tokens processed)	3.5471	3.5871
2C-10	9 layers, width 576; 75,000 sequences (76.8M tokens processed)	3.5451	3.5838
2C-11	8 layers, width 256	3.1054	3.1473
2C-12	8 layers, width 320	3.0288	3.0719
2C-13	8 layers, width 384	2.9737	3.0176
2C-14	8 layers, width 448	2.9218	2.9663
2C-15	8 layers, width 576	2.8600	2.8993
2C-16	8 layers, width 768	2.8048	2.8455
2C-17	4 layers, width 512	3.0120	3.0556
2C-18	5 layers, width 512	2.9752	3.0157
2C-19	6 layers, width 512	2.9344	2.9783
2C-20	7 layers, width 512	2.9084	2.9515
2C-21	9 layers, width 512	2.8662	2.9065
2C-22	12 layers, width 512	2.8323	2.8733

Problem 3(a): joint variation

ID	Changes from the standard recipe	Final train loss	Final validation loss
3A-01	model/data seed 1	2.8978	2.9285
3A-02	model/data seed 2	2.8833	2.9303
3A-03	model/data seed 3; A100	2.8799	2.9328

Problem 3: deterministic references

ID	Changes from the standard recipe	Final train loss	Final validation loss
3REF-01	deterministic execution	2.8858	2.9271
3REF-02	deterministic execution	2.8858	2.9271

Problem 3(b): isolated variation

ID	Changes from the standard recipe	Final train loss	Final validation loss
3B-01	model seed 1, data seed 42; deterministic execution	2.8819	2.9244
3B-02	model seed 0, data seed 42; deterministic execution	2.8873	2.9298
3B-03	model seed 2, data seed 42; deterministic execution	2.8902	2.9301
3B-04	model seed 42, data seed 0; deterministic execution	2.9330	2.9309
3B-05	model seed 42, data seed 1; deterministic execution	2.8939	2.9272
3B-06	model seed 42, data seed 2; deterministic execution	2.8762	2.9286

Problem 3(c): variability under interventions

ID	Changes from the standard recipe	Final train loss	Final validation loss
3C-01	dropout 0.2; model/data seed 0	3.1608	3.0639
3C-02	dropout 0.2; model/data seed 1	3.1148	3.0553
3C-03	dropout 0.2; model/data seed 2	3.0998	3.0609
3C-04	learning rate 0.027; model/data seed 0	3.0059	3.0009
3C-05	learning rate 0.027; model/data seed 2	2.9489	3.0019
3C-06	learning rate 0.027; model/data seed 1	2.9721	3.0114
3C-07	12 layers, width 768; model/data seed 0	2.7949	2.7884
3C-08	12 layers, width 768; model/data seed 1	2.7634	2.7967
3C-09	12 layers, width 768; model/data seed 2	2.7398	2.7880
3C-10	2,400,000 sequences (2.46B tokens); model/data seed 0	2.7355	2.7683
3C-11	2,400,000 sequences (2.46B tokens); model/data seed 1	2.7587	2.7700
3C-12	2,400,000 sequences (2.46B tokens); model/data seed 2	2.7035	2.7731

Problem 3(c): seeds 4–9

ID	Changes from the standard recipe	Final train loss	Final validation loss
3S-01	model/data seed 4	2.9759	2.9293
3S-02	batch 16; model/data seed 4	2.9721	2.9340
3S-03	learning rate 0.009; model/data seed 4	2.9842	2.9447
3S-04	batch 16; model/data seed 5	3.1215	2.9322
3S-05	model/data seed 5	2.9386	2.9323
3S-06	learning rate 0.009; model/data seed 5	2.9537	2.9470
3S-07	model/data seed 6	2.9094	2.9326
3S-08	batch 16; model/data seed 6	2.7982	2.9324
3S-09	learning rate 0.009; model/data seed 6	2.9323	2.9513
3S-10	model/data seed 7	2.8833	2.9357
3S-11	batch 16; model/data seed 7	2.8650	2.9417
3S-12	learning rate 0.009; model/data seed 7	2.9044	2.9579
3S-13	model/data seed 8	2.8737	2.9348
3S-14	learning rate 0.009; model/data seed 8	2.8797	2.9437
3S-15	batch 16; model/data seed 8	2.7214	2.9356
3S-16	model/data seed 9	2.8871	2.9334
3S-17	batch 16; model/data seed 9	2.9430	2.9318

Problem 4(a): one-token perturbation

ID	Changes from the standard recipe	Final train loss	Final validation loss
4A-01	deterministic execution; one training token changed	2.8858	2.9254
4A-02	deterministic execution	2.8859	2.9277

Problem 4(b): weight perturbation during training

Noise scale is a fraction of each parameter tensor's standard deviation. Training then continues to the end.

ID	Changes from the standard recipe	Final train loss	Final validation loss
4B-01	deterministic; perturb at step 100, scale 0.001	2.8863	2.9258
4B-02	deterministic; perturb at step 100, scale 0.01	2.8836	2.9261
4B-03	deterministic; perturb at step 100, scale 0.1	2.8845	2.9259

ID	Changes from the standard recipe	Final train loss	Final validation loss
4B-04	deterministic; perturb at step 2000, scale 0.001	2.8847	2.9262
4B-05	deterministic; perturb at step 2000, scale 0.01	2.8841	2.9264
4B-06	deterministic; perturb at step 2000, scale 0.1	2.8850	2.9265
4B-07	deterministic; perturb at step 5000, scale 0.001	2.8840	2.9262
4B-08	deterministic; perturb at step 5000, scale 0.01	2.8838	2.9258
4B-09	deterministic; perturb at step 5000, scale 0.1	2.8847	2.9268
4B-10	deterministic; perturb at step 8000, scale 0.001	2.8840	2.9262
4B-11	deterministic; perturb at step 8000, scale 0.01	2.8839	2.9262
4B-12	deterministic; perturb at step 8000, scale 0.1	2.8875	2.9301

Problem 5: diagnostic loss curves

ID	Changes from the standard recipe	Final train loss	Final validation loss
5-01	learning rate 0.0003	3.1116	3.1473